Specification note: the E-series architecture — equivalence-scale preferences, sequential fertility, and honest income risk

Fertility–housing project

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The change in one view. The E-series replaces the production model's two deepest provisional structures at once. Preferences lose their Stone–Geary intercepts — children price in through an expenditure-share shifter and a utility-level scale instead of committed bundles — and fertility becomes sequential: households try for a first child and one-child households may later try for a second, each attempt succeeding with an age-declining fecundity probability. Removing the intercepts removes the feasibility bind that capped income risk, so the income process is set externally at its literature value rather than tuned down to keep budget sets nonempty.

	Production (M5)	E-series
Child costs	floors $\bar{c}(n), \bar{h}(n)$ (5 params, SMM)	$\alpha(n,s)$, $\tilde{e}(n,s)$ (3 params, SMM)
Feasibility	bundle must be affordable; gate + censoring	every $y > 0$ feasible by construction
Income risk	$\sigma_z = 0.0645$ (feasibility-capped)	$\sigma_z = 0.20$, $\rho_z = 0.960$ (external)
Fertility	one-shot family-size choice	sequential attempts, fecundity π_j (external)
Free params / moments	14 / 15	12 / 15

Preferences. Period utility is

$$u(c,h;n,s) = \tilde{e}(n,s)\,\frac{\left(c^{\alpha(n,s)}h^{1-\alpha(n,s)}\right)^{1-\sigma}}{1-\sigma} + \psi n_k, \qquad \tilde{e} = 1 + \gamma_e n_k, \quad \alpha = \alpha_0 - (\delta_\alpha^{jump} + \delta_\alpha n_k)\mathbf{1}\{n_k > 0\},$$

with $\sigma = 2$ maintained, so $\tilde{e}$ coincides with the equivalence scale over the composite: a family needs $\tilde{e}$ times the childless composite for equal welfare. Textbook good-specific (Barten) scales are deliberately not used: under a Cobb–Douglas composite they cancel from the consumption–housing allocation, which would erase the child–housing demand channel the paper is about; the share shifter $\alpha(n,s)$ restores that channel in the form the consumption literature uses (demographic taste shifters), and $\tilde{e}$ carries the cost of children. With no intercepts, marginal utility diverges at zero and any positive income admits a feasible choice.

How the references handle the same object: Adda, Dustmann and Stevens (2017, JPE) *impose* the OECD-modified scale (1 + 0.5 per adult + 0.3 per child) on household consumption; Scholz, Seshadri and Khitatrakun (2006, JPE) impose the Citro–Michael scale $(A + 0.7K)^{0.7}$ in utility *and* in their means-tested transfer floor; Fernández-Villaverde and Krueger (2007, ReStat) deflate by the mean of six published scales; Blundell, Pistaferri and Saporta-Eksten (2016, AER) estimate demographic preference shifters inside their structural system; Borella, De Nardi and Yang (2023, ReStud) and De Nardi, Fella and Paz-Pardo (2025, ReStud) use divisor scales varying with marital status and children — the former imposing the Citro–Michael scale $(j + 0.7f)^{0.7}$, the latter imposing the OECD-modified weights (1, 0.5 per second adult, 0.3 per child). Guner, Kaygusuz and Ventura (2020, ReStud) explicitly reject utility scaling — but only to preserve balanced-growth consistency, a constraint our stationary economy does not face. Identification discipline for scales is the Pollak–Wales (1979, AER) critique: welfare-level scales are not identified from demand data alone. **Our approach differs from all of the above: we estimate γ_e inside the SMM** from housing, fertility, and wealth moments — and the estimate, $\gamma_e = 0.203$, implies +20.3% per one child and +40.5% for two, reproducing the OECD-modified child weights ($+20\%/+40\%$ on a couple base) that the literature imposes. Because our γ_e is identified by behavior through the model structure, this agreement is corroboration rather than circularity.

Fertility technology. The production model chooses completed family size in one event; the E-series prices time. Childless households choose {wait, try}; one-child households with the window open choose {stop, try for the second}; an attempt succeeds with probability $\pi_j = \max\{0, \min\{1, 1 - \omega_1 e^{\omega_2(a_j - 18)}\}\}$, zero from 45, and a second birth restarts the single child clock (per-child age tracking stays collapsed — the reason sequencing was originally dropped — at the small cost that the first child's maturation rides the youngest's clock). The schedule is *external*: biology identifies it, and pinning it keeps the fertility logit scale κ_f identified, since both objects smooth hazards. Calibration precedents and our fit are documented in the companion note (`fecundity_schedule_note_20260720`): Sommer (2016, JME) fits an exponential permanent-infertility CDF to Trussell–Wilson (1985), reaching ≈97% infertile at 45; de la Croix and Pommeret (2021, JET) fit a bounded logistic to Léridon's conception-by-duration table with a 4% natural-sterility level from Baudin, de la Croix and Gobbi (2015, AER). Our deployed $(\omega_1, \omega_2) = (0.02, 0.134)$ evaluates to $\pi = \{0.97, 0.90, 0.81, 0.62\}$ at ages 22/30/35/40 against Léridon's (2004) conceive-within-four-years anchors $\{0.96, 0.91, 0.84, 0.64\}$ — on the evidence to 1–3 points; the formal least-squares fit is a pending half-day refinement. New measured objects: attempt and realized birth hazards by age, first-birth age distribution, birth spacing via the flow progression from one to two children, and the decomposition of childlessness into chosen versus ran-out-of-clock.

Income process. Production carries $\sigma_z = 0.0645$ — a third of the PSID range — because the Stone–Geary bundle had to be affordable in every state (the July-18 frontier computation: infeasible above $\sigma_z \approx 0.09$ at the calibrated persistence). With no intercepts the constraint is gone, and the E-series imposes $(\rho_z, \sigma_z) = (0.960, 0.20)$ externally (the Sommer–Sullivan range), with no transfer floor needed.

Calibration accounting and status. Twelve free parameters (the five floors leave, $\delta_\alpha^{jump}, \delta_\alpha, \gamma_e$ enter) on the unchanged 15-moment income-disciplined target system; externals: $\theta_n = 0$, $\sigma = 2$, $\phi = 0.80$, the income process, and the fecundity schedule. The share shifters are identified by the family/childless housing and rooms moments; γ_e by fertility levels jointly with wealth accumulation. Current status (E2, July 20): loss 5.49 versus the production model's 9.04 on the same moments, cross-evaluator confirmed; old-age ownership and the estate distribution resolved; the open issues are young renters' liquid wealth (0.52 vs. 0.18) and the pending fecundity fit plus timing-moment data targets before paper-facing use.

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